

Suryalaxmi Ravianandan

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Objective

- Aspiring Data Scientist currently pursuing a Master's in Data Science with a specialization in machine learning and artificial intelligence at Friedrich-Alexander-University Erlangen-Nuremberg.
- Proficient in Python, core ML/DL libraries (Scikit-learn, Pandas, Matplotlib, Pydantic, Pytorch, Tensorflow), and specialized in developing knowledge-graph-driven solutions using Semantic Web technologies (RDF, SPARQL, RML).
- Motivated to join an agile team developing knowledge-graph-driven solutions for real-world business challenges.

Education

❑ **Master of Science(M.Sc.) in Data Science** 04 2022 - 09 2026

Major: Artificial Intelligence/Machine Learning
Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany
Note: 2.6

❑ **Bachelor of Engineering(B.E.) in Computer Science** 06 2006 - 05 2010

K.L.N college of Engineering, Tamilnadu, India
Bachelor Thesis: WiMAX - Worldwide Interoperability in Microwave Access
percentage: 69%

Skills

- **Core Data & AI/ML** : Python, SQL, Pandas, NumPy, Scikit-learn, Pydantic, PyTorch, TensorFlow, Matplotlib, Seaborn; machine learning (classification, regression), model evaluation (ROC-AUC, F1-score); large language models (LLMs) – prompt engineering, fine-tuning.
- **Platforms & Tools** : Git/GitHub, Jupyter Notebook, PyCharm, VS Code, LaTeX, Docker; FastAPI, REST APIs, MCP Server, JSON, JSON-LD
- **Knowledge Graphs** : RDF, RML, SPARQL, RDFLib, WoT Thing Descriptions; application domains: IoT (industrial shop floors)
- **Languages** : Tamil (native), English (fluent), German (B2)).

Project & Seminars

⇒ **Master thesis:** *"Automated Semantic IoT Data Integration for Flexible Shop Floors Using WoT Thing Descriptions"* - Sep 2026

- Developing an LLM-driven pipeline that parses WoT TDs to auto-generate RML mappings, transforming heterogeneous IoT data into RDF triples for ontology-flexible integration.

✓ **Project:** *Knowledge Graph Deconstruction*- Mar 2025

- Built an interpreter to parse RML mappings and convert RDF datasets to CSV, using SPARQL for targeted extraction from RDF sources to improve downstream analytics efficiency. <https://github.com/wintechis/ReversingRmlToCsv>
- Knowledge graphs are powerful but hard to analyze directly; this project simplifies semantic data handling for data scientists.

✓ **Seminar :** Data Governance

⇒ Foundation - *"Healthcare Robots - case study"* - Feb 2025

- Designed a data governance strategy for healthcare robotics, focusing on legal compliance, privacy, and ethical AI use.
- Integrated privacy-preserving methods and addressed tensions between fairness, security, and regulatory requirements (EU standards).

⇒ Application- *"Chatbot Training for Healthcare Applications "* - Sep 2025

- Exploring data governance challenges in healthcare chatbot development, focusing on data privacy, quality, and compliance.
- Implemented anonymization techniques to sensitive patient data, significantly reducing re-identification risks while preserving data utility for model training. <https://github.com/suravi73/Data-Governance-in-Healthcare---Robots-chatbot/tree/main/DG%20Application>

✓ **Master Seminar :** *Stochastic Compositional Gradient Descent* - July 2024

- Investigated SCGD algorithms for optimizing nonlinear functions involving expected values in stochastic environments.
- Implemented SCGD within a Markov decision process using Python, applying it to the Bellman residual for stochastic shortest path problems.

✓ **Seminar :** *Generalized Set Theory* - March 2024

- Investigated GST as a bridge between set theory and type theory; surveyed implications for foundational mathematics and formal systems.
- Exploring Generalized Set Theory (GST) as a bridge between set theory and type theory.
- Provides insights into mathematical foundations that could influence programming languages, logic, and AI theory.

Strengths

- **Data Analysis & Interpretation:** Skilled at extracting insights from complex data sets using advanced statistical and computational methods.
- **Ethical AI & Data Governance:** In-depth understanding of privacy, compliance, and fairness in AI systems, with practical experience in healthcare applications.
- **Adaptability & Continuous Learning:** Quick to adopt emerging tools, technologies, and methodologies in data science and AI
- **Collaborative & Independent Work Style:** Demonstrated both independent initiative and teamwork by executing an independent project and collaborating with a team during a Data Governance seminar.